

Exactness and Nuclearity of Uniform Roe Algebras: A Later-Literature Answer to Section 5.2 of arXiv:1101.1993

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Status

Literature already answered (affirmative), with limited scope. Sako's Theorem 1.1 answers the general exactness-versus-nuclearity question from Section 5.2 of Arzhantseva–Guentner–Spakula. The distinct question in Section 5.1 about equality of maximal and reduced uniform Roe algebras is not claimed here.

Source question

For a bounded-geometry metric space X , let $C_u^*(X)$ be the norm closure on $\ell_2(X)$ of the finite-propagation matrices with uniformly bounded entries. On PDF page 13 of arXiv:1101.1993, the authors recall that

$$X \text{ has Property A} \iff C_u^*(X) \text{ is nuclear}$$

and ask whether, for arbitrary bounded-geometry X ,

$$C_u^*(X) \text{ exact} \implies C_u^*(X) \text{ nuclear.}$$

They also ask whether the uniform Roe algebra of their particular coarsely-Hilbert-embeddable, non-Property-A box space is exact, before noting a box-space argument that already makes it nonexact.

Explicit answering theorem

Sako, arXiv:1212.5900, Theorem 1.1 (PDF page 1), proves that for every metric space X with bounded geometry the following are equivalent:

1. X has Property A;
2. $C_u^*(X)$ is nuclear;
3. $C_u^*(X)$ is exact;
4. $C_u^*(X)$ is locally reflexive;
5. X has the operator norm localization property.

In particular, exactness implies nuclearity, so the source’s general question has an affirmative answer. Sako’s abstract and introduction explicitly say that exactness for general bounded-geometry spaces had been open until the draft was announced, making this an exact literature answer rather than a coincidental theorem match.

For the source’s box space

$$X = \bigsqcup_n \text{Cay}(F_2/\Gamma_n), \quad \Gamma_{n+1} = \langle g^2 : g \in \Gamma_n \rangle,$$

the source proves failure of Property A. Sako’s equivalence therefore gives that $C_u^*(X)$ is neither nuclear nor exact.

Scope limitation

Section 5.1 separately asks whether

$$C_{u,\max}^*(X) \longrightarrow C_u^*(X)$$

is an isomorphism for this same box space. Coarse Hilbert embeddability gives an isomorphism on K-theory, not injectivity of the algebra map. Willett’s arXiv:1504.05615, Remark 3.5, still describes the coarse-space equality problem as open and explains that his nonamenable groupoid with equal maximal and reduced algebras does not resolve the coarse case. A bounded search through 2026-08-11 found no later exact answer for the iterated-square box space. The present run records a separate serious attempt, but promotes no claim on that companion question.

Search evidence

The statements were checked in the official PDFs and parsed sources of the three cited arXiv papers. Searches for the exact question, titles, authors, maximal uniform Roe algebras, box spaces, and Property A were run in the local corpus, arXiv, and bounded citation neighborhoods. Later hits extend the K-theory result or study adjacent groupoid phenomena; none found in this audit settles the remaining isomorphism question.

References

- [1] G. Arzhantseva, E. Guentner, and J. Spakula, *Coarse non-amenability and coarse embeddings*, arXiv:1101.1993; *Geom. Funct. Anal.* 22 (2012), 22–36.
- [2] H. Sako, *Finite-dimensional approximation properties for uniform Roe algebras*, arXiv:1212.5900, Theorem 1.1.
- [3] R. Willett, *A non-amenable groupoid whose maximal and reduced C^* -algebras are the same*, arXiv:1504.05615, Remark 3.5.